

BOOK FIVE- A must read - how to
make your backhand into a weapon

ONE MINUTE TENNIS **BACKHAND SOLUTION**



Improve Faster- Play Smarter
Stephen James Bourne

ONE MINUTE TENNIS
BACKHAND SOLUTION

Stephen James Bourne

One-minute tennis backhand solution

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How to Use this Guide

There are many factors that make the difference between winning and losing in tennis.

Fitness, athletic ability, insufficient practice time, technique, and mental strength under pressure.

For the vast majority of players, **the biggest limitation is 'technique'.**

One Minute Tennis Solutions concentrate on the technical aspect of the game and are beneficial for players of all levels.

I have included detailed information on the biomechanical principles of each stroke, the anatomy, and the physics involved.



If you have suffered from injuries, then understanding what part of your body is involved in the different stages of the strokes, can be a great way of preventing future problems.

Accompanied by simple and effective 'One Minute Solutions.'

How to Use this Guide

Science helps you to understand the stroke.

One Minute Solution helps you improve your stroke.

At the end of each One Minute Solution, there is a link to the same lesson in video format.



These solutions work almost instantly.

To benefit from the advice, take your time to feel and visualize the shapes and movements.

At first, use shadow strokes and slow controlled practice and then you will see your real tennis potential emerge.

As well as using the One Minute Solutions you can also study the precise science and biomechanics of each aspect of the strokes to understand how the modern tennis game really works.

Visualize, Feel & Understand Tennis

The One Minute System

"Hi, I'm Steve, welcome to my One Minute Tennis System."

For many years I was privileged to run a very successful tennis academy.

I worked with high-level players, world-class juniors and recreational players of all ages and levels.

During this time I studied anatomy and biomechanics and developed a deep understanding of the scientific principles of tennis.

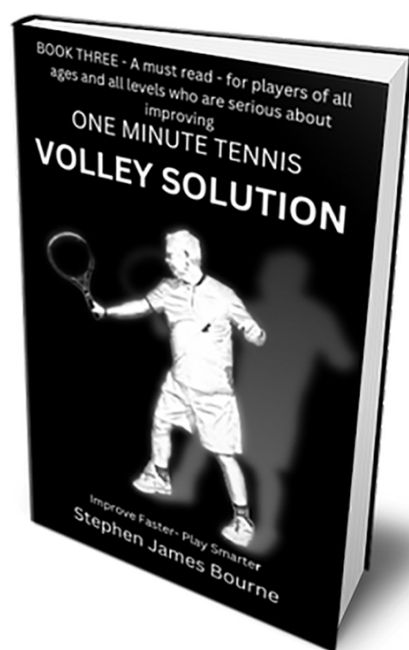
Utilizing my knowledge of the science and physics of the game I created a unique teaching system based on feel, sensation, and imagery.

This system proved to be incredibly successful with players of all levels. Who improved faster than they ever thought possible.

The Language of the Mind is Words- The Language of the Body is Feeling!

Essential teaching is the precise moment when a player actually understands a new concept and is able to apply it into their game.

This information should be explained in one minute or less.



The One Minute System

Traditional teaching methods label us as either a

Visual Learner

Audible Learner

Kinaesthetic Learner

This is often called VAK teaching and is how tennis has been taught throughout history.

People often say

“I have to see this written down”

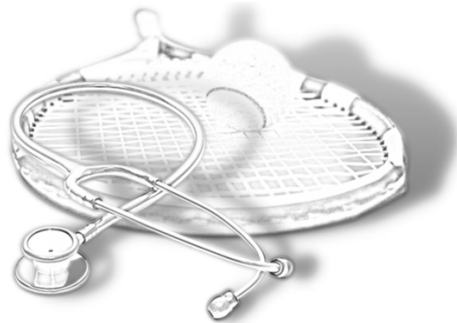
“I have to see things” or

“I have to feel this”

Of course, they are right, seeing, feeling, and touching are essential for learning practical skills.

But ... I created a better way, helping players to learn incredibly quickly, in an almost superhuman fashion. To achieve this we communicate to all of the learning senses at the same time.

This is called *Blended Learning or Blended Teaching*.



1. Feel the movement.



2. See the movement.



3. Understand the movement.



Backhand Grip Biomechanics

1. Grip Analysis

To find a perfect backhand grip, place the base knuckle of your index finger on the third bevel of the racket handle.

This grip position allows for a semi-western orientation of your hand, enhancing your ability to generate topspin and power while maintaining control and stability.:



Backhand Grip Biomechanics

2. Joint Mechanics:

With the Eastern or Semi-Western backhand grip, your wrist assumes a firm and neutral position.

This enables you to maintain stability and control during the swing, especially when making contact with the ball.

A stable wrist, also minimizes the risk of injuries associated with excessive wrist flexion or extension.



3. Muscle Activation:

Your execution of the efficient backhand grip relies on specific muscles for producing power and control during the stroke.

The pronator quadratus, flexor carpi ulnaris, and extensor carpi radialis muscles play key roles in stabilizing your wrist and forearm during the swing.

Backhand Grip Biomechanics

4. Kinematic Chain:

To optimize the biomechanics of the grip, your body must coordinate an efficient kinematic chain.

This involves synchronized movements of your lower body, trunk, and upper extremities.

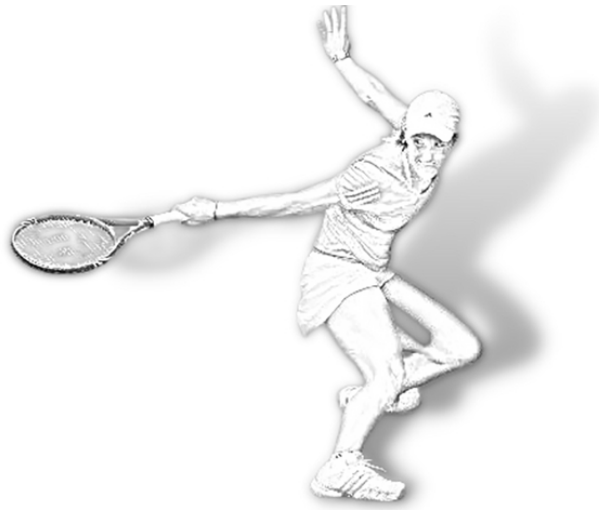
Proper sequencing of these movements is critical for generating power and transferring energy from the ground up through your body and into the racket.



Backhand Grip Biomechanics

5. Body Rotation:

Effective body rotation plays a significant role in maximizing the potential of the backhand grip.



Initiate the swing with a rotational movement in your hips and shoulders. This gathers and stores elastic energy in your muscles and tendons.

During the forward swing, this stored energy is released smoothly via the neutral wrist of your Eastern or Semi-Western grip. This results in a natural increase of racket head speed.

Backhand Grip Biomechanics



6. Angular Momentum:

The physics behind the correct backhand grip takes advantage of the principle of angular momentum.

As you swing the racket in an arc, angular momentum is created, resulting in increased racket head speed and

consequently more power and control.

Backhand Grip Biomechanics

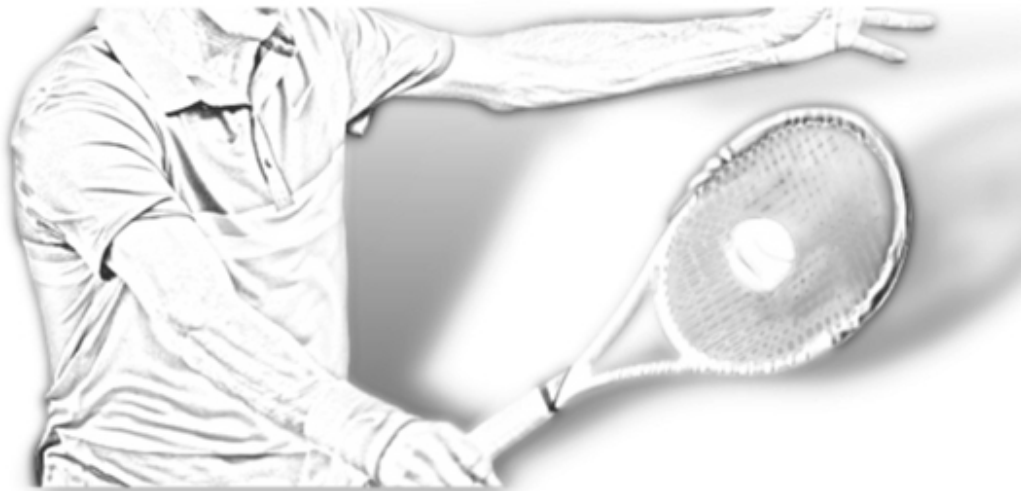
7. Follow-Through:

This is a crucial aspect of grip biomechanics.

A complete and balanced follow-through ensures optimal energy transfer from your swing to the ball and prevents strain on your arm as the racket decelerates.

In summary, the Eastern or Semi Western backhand grip in tennis involves a complex interplay of joint mechanics, muscle activation, and kinematic chain sequencing.

Understanding and optimizing these biomechanical principles, helps you to avoid injury and improve the quality of your stroke.

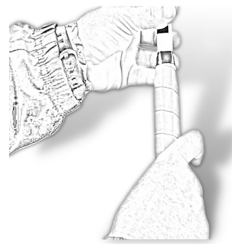


Backhand Grip Physics

When you use the backhand grip in tennis, you engage various principles of physics.

By placing the base knuckle of your index finger on the third bevel of the racket handle, you gain a grip that allows for a relatively neutral wrist position, giving you good control and versatility on both topspin and flat shots.

As you swing the racket in a circular motion during your backhand stroke, you generate angular momentum.



The grip mechanics enable you to rotate the racket around its longitudinal axis easily, increasing your racket head speed and empowering you to hit powerful shots.

During your swing, centripetal force acts on the racket, pulling it towards the center of rotation (your hand).

Thanks to the stable contact point provided by the Eastern or semi Western grip's positioning, you can maintain control over the racket despite the centrifugal force generated during the swing.

Backhand Grip Physics

Friction comes into play when the strings of your racket make contact with the ball.

The closed racket face of the Eastern or Semi-Western grip allows the strings to grip the ball effectively, resulting in greater friction during the collision.

This friction causes the ball to roll forward on the strings which creates topspin. Resulting in better depth and control. The energy transfer from your body to the racket, then to the ball, determines the speed and power of your shot.

As you step into contact and uncoil your body, *potential energy stored in your muscles converts into kinetic energy.*

This kinetic energy, is then transferred to the racket, and upon impact transferred to the ball.

The correct grip facilitates efficient energy transfer, enabling more powerful ball striking.



Backhand Grip Physics

After contact with the ball, a smooth follow-through is essential, as this is responsible for redistributing the angular momentum initially built during your backswing and forward swing.



Backhand Grip Physics

The closed racket face at contact with the correct Eastern or Semi Western grip contributes to a lower launch angle for the ball, helping it clear the net with sufficient margin.

By focusing on proper mechanics and adjusting your swing, you can control the height of your shot and reduce the chances of hitting the ball into the net.



In summary, using the correct backhand grip involves applying principles of *angular momentum*, *centripetal force*, *friction*, and *energy transfer*.

Backhand Grip Solution

There are many variations of backhand grips, but they can be divided into just two categories:

Good Grips that function and Bad Grips that do not.

If your wrist is level or ahead of the ball when it makes contact with the racket strings, then the grip is inherently wrong.

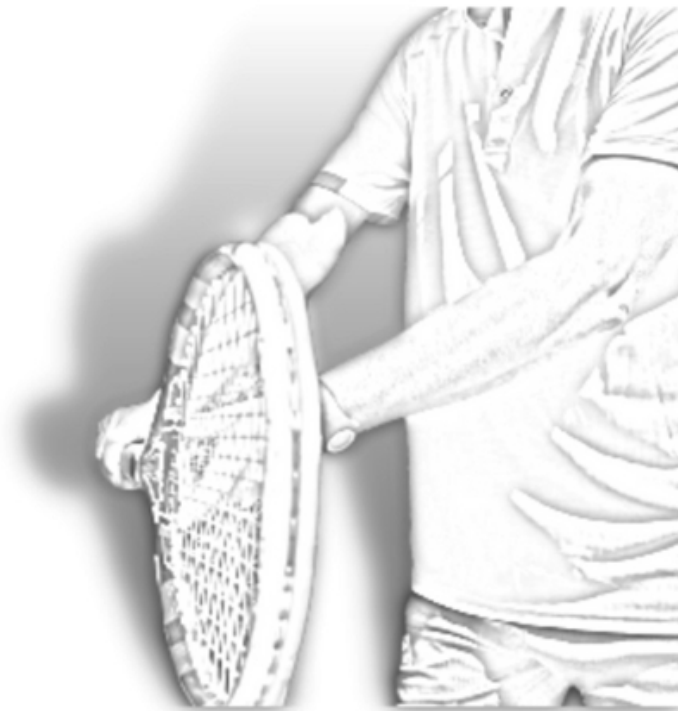


This grip will be weak and could potentially cause injury to your wrist and elbow.

Backhand Grip Solution

On the other hand, if your wrist is behind the ball when it makes contact with the racket strings, then the grip is inherently correct.

It will be strong, and your arm will be protected from stress on impact.



Some players turn the racket a long way and have a Semi-Western backhand grip with the wrist positioned far behind the strings.

Others make a smaller turn, and the wrist is positioned just slightly behind the contact in the Eastern Grip.

Backhand Grip Solution

To find the right grip for you, follow these steps:

- 1. Place your racket under your left arm with the handle facing forwards in front of you.**



Backhand Grip Solution

2. Now bring your right hand above the handle of your racket and take hold of the racket.



Backhand Grip Solution

3. You will now be holding your racket in a backhand grip and are very likely to be in the correct grip for you.

Remember, both Semi-Western or Eastern grips are correct. Using this method to find your grip will help you to determine which is the natural grip for your backhand





Backhand Grip Solution

To help you to find this grip with speed and precision.

Begin your one-handed backhand with the correct ready position.

As with all groundstrokes, you should have a wide base with the heels slightly raised and your knees bent.



Try to keep your head slightly ahead of your knees, as this will ensure an athletic stance and the ability to move swiftly in any direction.

Backhand Grip Solution



Your left hand should be the dominant hand with your fingers spread and the grip on the racket quite firm.

Your right hand should be relaxed and very loose.

Backhand Grip Solution



As the left hand leads the take back, the relaxed right hand should allow the racket to turn within the fingers.

Your racket will naturally turn into the correct grip by the end of the take-back.

YouTube: https://youtu.be/-aRcX_SXRsq

Backhand Footwork Biomechanics

1. Stance and Balance:

Achieving a solid and stable stance is fundamental.

A slight knee flexion and hip abduction, creates a stable base. Optimizing your ability to transfer force efficiently through the lower extremities.

Balance is essential for rapid changes in direction and weight transfer during the stroke and this knee flexion and hip abduction will ensure you are balanced with an athletic position.



Backhand Footwork Biomechanics

2. Weight Transfer and Center of Mass:

Efficient weight transfer from your back foot to your front foot is essential for generating power.

The shift in weight creates a forward momentum, which is crucial for generating racket head speed. Your center of mass should move in the direction of the shot to maintain balance and control.



3. Pivot and Angular Momentum:

The pivot on the back foot, enables rotation of your hips and shoulders.

This rotation generates angular momentum, allowing you to store potential energy in your core that can be transferred as kinetic energy to your racket.

Backhand Footwork Biomechanics

4. Step-In and Lateral Push-Off:

The step-in involves a controlled movement of your front foot towards the ball.

This action is accomplished by a lateral push-off with your back foot.

This increases the time of contact with the ground and enhances ground reaction forces.

A forceful lateral push-off generates more power.



5. Hip and Shoulder Separation:

During the preparation phase, your hips and shoulders rotate in opposite directions.

This separation increases the stretch of your core muscles, enabling a more forceful rotational movement.

Backhand Footwork Biomechanics

6. Contact Point and Kinetic Linkage:

Striking the ball in front of your body and at waist height optimizes the kinetic linkage from your lower body to your upper body and your racket.



7. Follow-Through and Deceleration:

A complete follow-through involves the continuation of the swing after contact with the ball.

A proper follow-through is essential as it allows for the deceleration of the racket in a controlled manner, reducing the risk of injury and helping to maintain balance.

Backhand Footwork Anatomy

1. Stance and Weight Distribution:

Begin with an open stance facing the net.

Ensure your feet are approximately shoulder-width apart, and your knees are slightly flexed to facilitate quick movements.

Your weight should be evenly distributed, allowing you to move smoothly in any direction.



Backhand Footwork Anatomy

2. Plantar Fascia and Foot Arch Support:

The plantar fascia is a dense band of connective tissue running along the bottom of your foot and plays a vital role in foot arch support and shock absorption.

This structure helps maintain the foot's arch, providing stability, and minimizing stress during weight transfer and pivoting movements.



3. Ankle Joint Stability:

The ankle joint, consisting of the *tibia, fibula, and talus bones*, is responsible for supporting your body weight and facilitating movement during the backhand stroke.

Ligaments such as the *lateral ligament complex* help stabilize your ankle, preventing excessive rolling or sprains during quick lateral movements.

Backhand Footwork Anatomy



4. Metatarsophalangeal (MTP) Joints and Toe Propulsion:

The MTP joints, located between the *metatarsal bones and phalanges* of your toes, are involved in the toe-propulsion during the push-off phase.

As you step forward and push off with your back foot, your toes contribute to generating forward momentum, enabling powerful weight transfer and shot execution.

Backhand Footwork Anatomy

5. Achilles Tendon and Calf Muscles:

Your *Achilles tendon*, the largest and strongest tendon in the body, connects your calf muscles (*gastrocnemius and soleus*) to your heel bone (*calcaneus*).

This tendon is crucial for transmitting forces from your calf muscles to your foot during the jumping, pivoting, and push-off movements all of which are essential for a strong backhand.

6. Hip Joint Mobility and Rotation:

Your hip joint allows for a wide range of motion and is pivotal in generating power during the stroke.

Correct hip rotation facilitates the coil and uncoil action, enabling you to store and transfer energy effectively from the lower body to the upper body.

Backhand Footwork Anatomy

7. Knee Joint Flexion and Stability:

Your knee joint, is a hinge joint connecting your *femur to your tibia and* undergoes flexion during the weight transfer and shot execution phases of your stroke.

Proper knee stability and controlled flexion are essential for efficient force transfer and injury prevention.



8. Lumbar Spine and Core Activation:

Your *lumbar spine*, located in the lower back, is involved in the twisting motion during the backhand stroke.

Engaging these core muscles, including the *Transverse abdominis and obliques*, is essential for stabilizing your spine, facilitating rotation, and generating power.

Backhand Footwork Anatomy



9. Neuromuscular Co-ordination and Proprioception:

Successful execution of backhand footwork relies on the interplay of *neuromuscular coordination and proprioception*.

Your brain communicates with your muscles and joints to precisely time movements, adjust balance, and maintain correct positioning on the court.

Backhand Footwork Physics

1. Weight Transfer:

The *law of conservation of momentum* plays a role in weight transfer during the backhand.

As you shift your weight onto your back foot (non-dominant side), *you conserve momentum, storing potential energy in your body.*

This stored energy is then transferred to your front foot (dominant side) during the push-off phase.

2. Pushing Off:

The principle of *Newton's third law of motion*, also known as the law of action and reaction, comes into play when you push off from your back foot.

As you exert a force against the ground, the ground exerts an equal and opposite reaction force, propelling you forward with an action-reaction effect

Backhand Footwork Physics

3. Angular Momentum:

Angular momentum is governed by *the conservation of angular momentum law*.

As you rotate your body during the backhand, the total angular momentum remains constant unless acted upon by an external torque.

By coiling your body and conserving angular momentum, you can unleash rotational power during the shot.

4. Lower Body Alignment:

The law of gravity dictates that your body's center of mass must be within your base of support to maintain stability.

Keeping your feet shoulder-width apart using a slightly open stance ensures that your center of mass stays well-balanced, reducing the risk of falling during quick movements.

Backhand Footwork Physics

5. Balance and Stability:

The concept of center of mass and balance is governed by the *law of gravity*.

When you move and shift your weight during the backhand footwork, your center of mass stays aligned with your base, enabling and encouraging you to maintain balance and stability.



Backhand Footwork Physics



6. Deceleration and Control:

Newton's second law of motion explains deceleration during the backhand footwork.

Applying a braking force results in a negative acceleration(deceleration), allowing you to slow down your forward momentum effectively.

This controlled deceleration gives you more time to set up for your stroke.

7. Transferring Force:

Force transfer in the footwork involves the principle of impulse-momentum theorem.

The impulse applied by your leg muscles during the push-off phase changes your momentum, transferring the energy through your kinetic chain to your upper body and racket.

Backhand Footwork Solution

The footwork on the backhand can be confusing. It's important that your right foot steps across your body before you make contact with the ball. **This, however, should be the last step, not the first step.**

The problem is that most players actually step across with their right foot as their first step to the ball. When your right foot is the first step, you become trapped with your weight anchoring you in this position.



If the ball is too far in front, you have to lean forwards.
If the ball is too far to the side, you have to lean to the side
if the ball is too close, then you are jammed and often can only make a weak slice.

So the natural and traditionally taught footwork is ineffective and problematic.

Backhand Footwork Solution

Begin by stepping to the ball with your left foot, not your right foot, and then let the right foot follow as a secondary step, almost as a part of the stroke.

This flowing movement will help you adjust to the ball's position effectively.



If the ball is too far in front.

Then after the initial step with your left foot, you will naturally take a large step forward with your right foot and therefore you will be perfectly positioned for the short ball.

Backhand Footwork Solution

If the ball is too far to your left.

Then after the initial step with your left foot, you will naturally take a large step to your left with your right foot and therefore you will be perfectly positioned for the wide ball.

Even if the ball is too close.

You play an open stance backhand.

This time by simply doing nothing after the first step with your left foot, you can still execute the stroke superbly from an open stance.

By leading with your left foot first and using your right foot as an adjusting step, you'll find yourself in a good position to handle the ball almost every single time.



Backhand Footwork Solution

Power From the Ground Up

To generate more power, focus on landing your lead foot, the right foot, with your heel first and your toes second.

This biomechanically proven movement allows energy to flow from the ground upwards, utilizing kinetic chains effectively.

Creating a natural flow of power through the conservation of energy.



When you step with your heel first and your toes second, the energy will travel through your foot, ankle and knee into your stroke.

Backhand Footwork Solution

This will result in easier power and more topspin in your backhand shots.

If there isn't enough time to execute a heel-to-toe landing, it's not a problem because that means you don't need the extra power at that moment.

When you do have time, using the heel-to-toe movement will add speed to the ball. On the other hand, if time is limited, you can still rely on the ball's existing speed.



So, remember, step across with your left foot first, followed by the right foot in a flowing and dynamic manner.

Use the heel-to-toe landing to add more power and maintain perfect spacing.

YouTube: <https://youtu.be/LjCnEkzLyu8>

Backhand Swing Path Biomechanics

1. Shoulder Rotation and Scapular Control:

During the upswing, your shoulder joint undergoes external rotation, facilitated by the *deltoid and rotator cuff muscles*.

Your scapular is controlled by

Your *rhomboids and trapezius muscles*. Correct positioning of these muscles creates a stable base for the stroke.



Backhand Swing Path Biomechanics

2. Forearm Supination and Pronation:

As you take the racket back for the upswing.

Forearm supination prepares your racket face for an open position.

During the upswing phase, *forearm pronation* is initiated to adjust the racket face angle in preparation for the contact point.



Backhand Swing Path Biomechanics

3. Elbow Flexion and Extension:

The upswing of your racket involves elbow flexion, controlled by the *biceps brachii*, which raises your racket toward the ball.

The subsequent extension, assisted by the *triceps brachii*, allows for acceleration of the racket head as it approaches the contact point.



Backhand Swing Path Biomechanics

4. Wrist Extension and Radial Deviation:

During the upswing, wrist extension helps position your racket for an optimal contact point with the ball.

Radial deviation assists in fine-tuning the angle of your racket face, aiding precision and control.

5. Core Rotation and Hip Pivot:

Generating power in the upswing requires coordinated core rotation.

The *obliques and transverse abdominis* contribute to twisting your trunk, while the hip pivot enables smooth and efficient transfer of energy from the lower body to the upper body.

Backhand Swing Path Biomechanics

6. Weight Transfer and Ballistic Movement:

As you initiate the upswing there is a transfer of weight from your back foot to your front foot.

This transfer, facilitates a ballistic movement of your racket, creating momentum and increasing potential energy.



Backhand Swing Path Biomechanics

7. Kinetic Linkage and Segmental Sequencing:

The upswing utilizes *kinetic linkage and segmental sequencing* to optimize energy transfer.

Sequential activation of the body segments from the ground up creates an effective transfer of force, from your legs and core to your arm and racket.



8. Visual Focus and Anticipation:

During the upswing of the racket, maintain visual focus on the ball to time your stroke accurately.

Extrapolate the ball's trajectory, adjusting your body positioning and racket accordingly.

Backhand Swing Path Biomechanics

9. Racket Lag and Stretch-Shortening Cycle:

A proper upswing involves creating racket lag by delaying the forearm pronation.

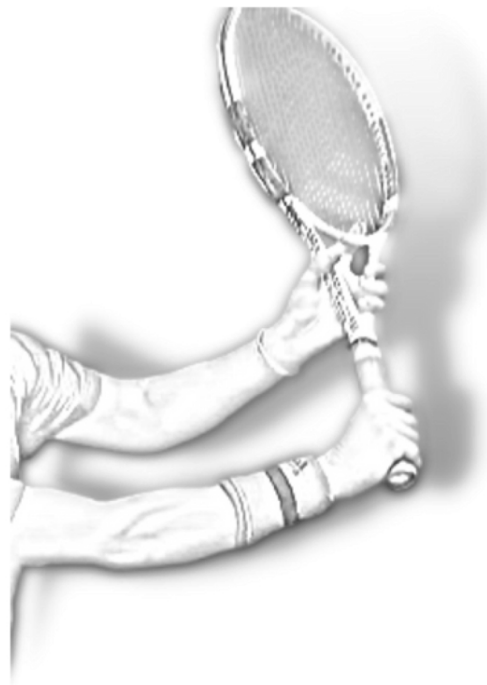
This motion generates stretch in the muscles and tendons, which contributes to the stretch-shortening cycle, enhancing racket head speed during the forward swing.

10. Breathing and Muscle Activation Timing:

Good breathing patterns and correct muscle activation timing are essential during the upswing.

This is much easier to achieve if you are relaxed.

Exhaling during the exertion phase can help enhance muscle activation and coordination, improving the fluidity of your movement.

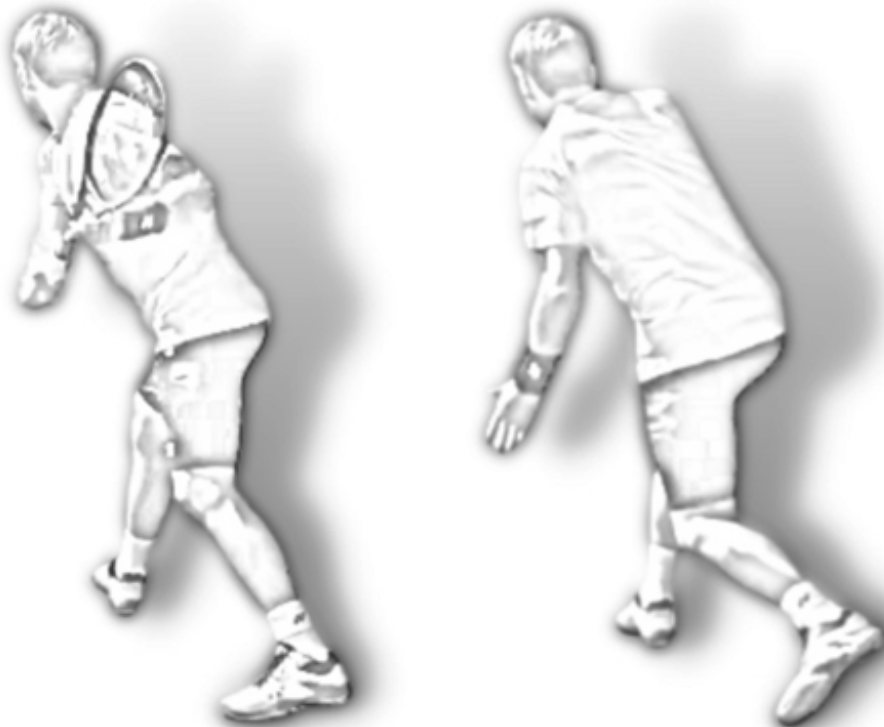


Backhand Swing Path Anatomy

1. Shoulder Muscles:

During the upswing, your shoulder muscles, including the *deltoids and rotator cuff* muscles, initiate and control the rotation of your arm.

The *deltoids* help lift the arm, while the rotator cuff muscles stabilize the shoulder joint during this dynamic movement.



Backhand Swing Path Anatomy

2. Forearm Muscles:

The muscles of your forearm, such as the wrist extensors and flexors, are highly active during the upswing.

These muscles work to position your racket correctly, encouraging precise control and preparation for the contact point.

3. Upper Arm Muscles:

Your *biceps brachii* and *triceps brachii* play a crucial role in the upswing.

The biceps, help flex your elbow bringing your racket closer to your body, while your triceps assist in extending your elbow during the preparatory phase of the stroke.



Backhand Swing Path Anatomy

4. Elbow Joint and Ligaments:

As your elbow undergoes flexion during the upswing, the *medial and lateral collateral ligaments* provide stability to your joint.

Correct elbow positioning and flexibility are essential for generating racket head speed and achieving optimal racket position.

5. Core Muscles:

Your core muscles, including the abdominals and obliques, contribute to the rotation of your trunk during the upswing.

A strong and coordinated core, facilitates the transfer of energy from your lower body to your upper body in preparation for the execution of your stroke.

Backhand Swing Path Anatomy

6. Hip and Pelvis Movement:

The backhand upswing, requires a controlled movement of your hips and pelvis.

Proper hip rotation and pelvic stability, facilitate the coil and uncoil action.

This enables the generation of torque and angular momentum, which contributes to your strokes power and accuracy.



Backhand Swing Path Anatomy

7. Scapular Stabilizers:

The muscles that stabilize your *scapula* (shoulder blade), such as the *rhomboids and trapezius*, help maintain correct shoulder positioning and *scapular* control during the upswing of your stroke.

This stability is essential for efficient and controlled arm movements.



8. Visual Focus and Neck Muscles:

Intense visual focus on the incoming ball during your upswing is crucial for timing and preparation.

The muscles of your neck and cervical spine stabilize your head and maintain the visual focus, ensuring proper stroke anticipation and execution.

Backhand Swing Path Physics

1. Newton's First Law of Motion (Law of Inertia):

Remember that your racket and arm have inertia. This means that they will stay at rest or in motion unless you apply an external force.

You need to exert muscular force to start the upswing and accelerate the racket.

2. Newton's Second Law of Motion (Law of Acceleration):

The force that you apply during the upswing directly affects your racket's acceleration.

The heavier your racket, the more force is needed to achieve the desired speed.

3. Newton's Third Law of Motion (Action-Reaction):

As you exert force on your racket during the upswing, your racket exerts an equal and opposite force on your arm.

This action-reaction enables the transfer of energy to your racket.

Backhand Swing Path Physics

4. Law of Conservation of Angular Momentum:

During the upswing, as you bring your racket closer to your body, your racket's rotational speed increases to conserve angular momentum.

This results in a rapidly accelerated upswing.

5. Law of Conservation of Energy:

Your muscular effort increases your racket's kinetic energy during the upswing.

This energy is then converted into other forms during the follow-through and execution of your stroke.

Backhand Swing Path Physics

6. Law of Torque (Moment of Force):

You apply torque to your racket during the upswing, causing it to rotate around your shoulder joint.

The amount of torque depends on the force you apply and the distance from your shoulder joint.

This is why tennis players are getting taller with longer limbs. These long levers create greater torque.



Backhand Swing Path Physics

7. Law of Projectile Motion:

As you swing your racket upwards, it follows a projectile motion trajectory.

The initial angle, your racket's velocity, and gravity, influence the trajectory.

This determines the position of your racket at the point of contact.



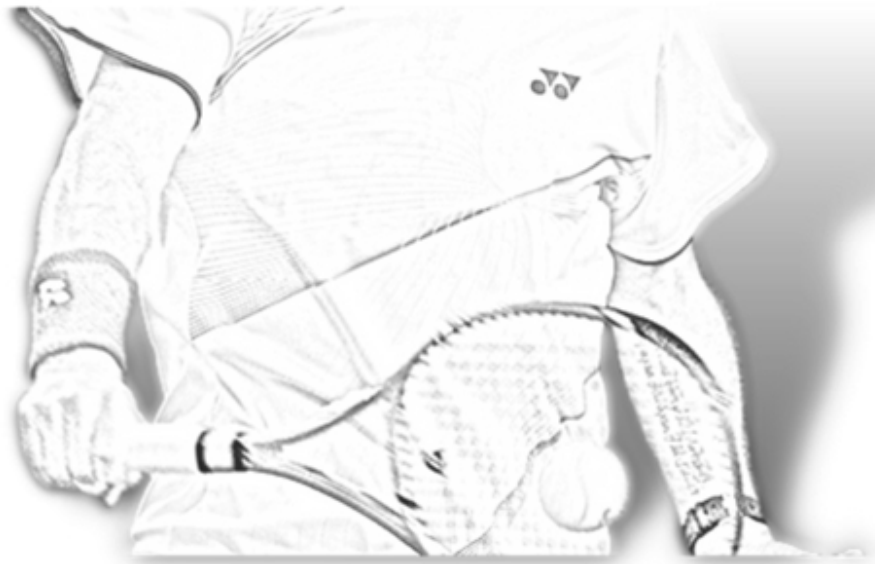
Backhand Swing Path Physics

8. Law of Elasticity:

When your racket contacts the ball, both objects experience deformation due to elasticity.

The efficiency of energy transfer during the collision depends on the material's elasticity.

String type and string tension will greatly affect the degree and type of deformation and how much energy is released into the ball.



Backhand Swing Path Physics

9. Law of Conservation of Momentum:

At the contact point, momentum is conserved.

The momentum of your racket and ball before the collision equals the combined momentum after the collision.

This transfers momentum to the ball, propelling it in the opposite direction with a force equal to energy of the swing.

Backhand Swing Path Solution

The swing path of the backhand involves a low to high and inside-out movement.

This stroke and especially the swing path can be quite confusing because it's easy to imitate the swing path of the best players in the world and still produce a slow, weak, and pushy stroke.

Let's explore the small but essential secrets to achieving a powerful backhand stroke.

At the end of your backswing, ensure that your left hand is positioned above your right hand, with your racket head at approximately shoulder or head height.



Backhand Swing Path Solution

In one swift smooth and rhythmical movement, reverse the racket head from above your hands to below your hands.

Then immediately begin the upswing towards the ball. So your racket head is moving from high, to low and then recoiling into the swing and into contact with the ball.



Backhand Swing Path Solution

To accomplish this, it's crucial to focus on the specific positions and relationship of your hands and your racket.



Backhand Swing Path Solution

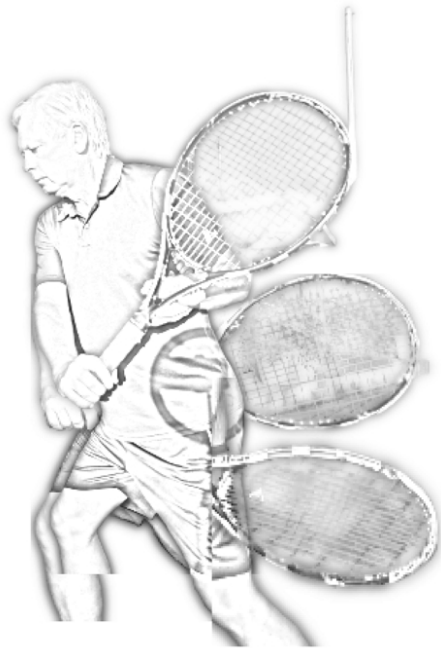


Backhand Swing Path Solution

As you lower your racket in a smooth and rhythmical manner, your left hand moves below your right hand. Then your hands must immediately separate and the upswing to the ball begins.

At first, after reversing the position of your racket and hands, you will have to consciously make the upswing. Then, after a few repetitions, this flowing movement will naturally transition into the upswing.

This swing should be fast, and the acceleration should be smooth.



Backhand Swing Path Solution

Applying the left-hand up, right-hand down, to left-hand down, right-hand up principle is essential.

The swift reversal of your hand positions creates tremendous momentum in your racket head, thereby immensely increasing the power of the stroke.

It's quite easy to develop the habit of *pushing* the one-handed backhand, leading to a loss of power and control and resulting in unstable contact with the ball. *(Remember that habits are easy to form and hard to break, so we need to create perfect habits).*

If you feel that you are falling into this pushing tendency, here is a great solution:



Backhand Swing Path Solution

Feel for the end of your backswing and at that final moment when your racket head is dropping, pull your racket upwards and into contact with the ball.



There is a distinct difference between pushing and pulling.

If you are pushing your racket, it feels like you are overcoming a force in front of you that is stopping your racket from moving forward.

If you are pulling your racket, it feels like you are overcoming a force from behind you that is holding your racket back.

Backhand Swing Path Solution



By reversing your hands, dropping your left hand below your right hand, and then pulling your racket forwards and up, you will achieve a powerful, energized, and smooth swing. This will result in successful contact with the ball and a natural powerful topspin drive.

YouTube: <https://youtu.be/dovjc0-tj0k> and
<https://youtu.be/-IXPxPazT7U>

Backhand Contact Point Biomechanics

1. Angular Motion and Joint Mechanics:

At the contact point, your body undergoes angular motion as you swing your racket.

The shoulder joint, elbow joint, and wrist joint work together to produce a coordinated movement, generating racket head speed and controlling the direction of your stroke.



Backhand Contact Point Biomechanics

2. Muscle Activation and Force Production:

The muscles of your upper extremities play a crucial role in generating force at the contact point.

The muscles of the forearm, such as the *wrist extensors* and *flexors*, contribute to wrist stability and control, while the *rotator cuff muscles* of your shoulder aid in the acceleration and deceleration of your racket.

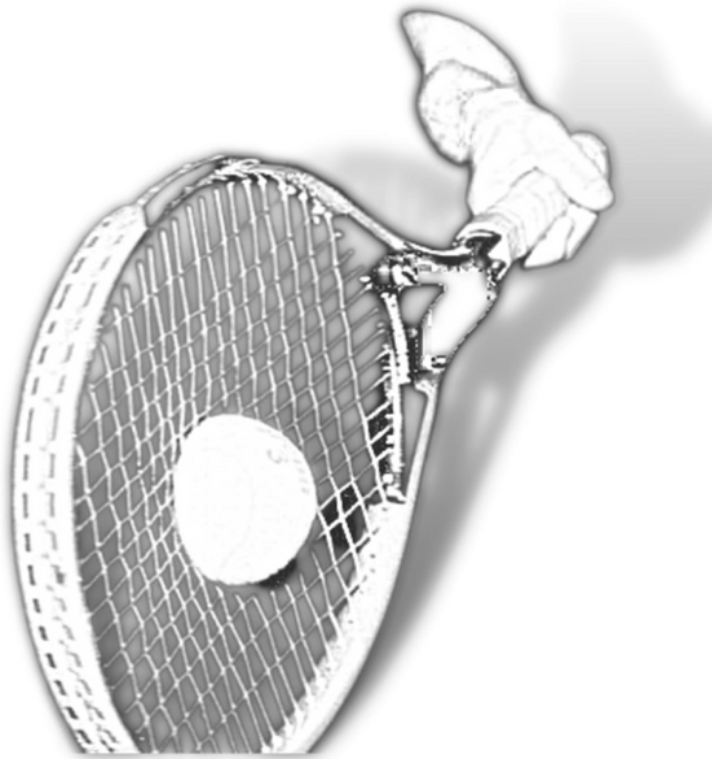


3. Translational Motion of the Ball:

The contact point initiates the translational motion of the ball.

Depending on the angle and speed of your racket, the ball's trajectory, speed, and spin are determined during this critical phase.

Backhand Contact Point Biomechanics



4. Ball Compression and Energy Transfer:

As your racket makes contact with the ball, both objects undergo deformation and compression.

The ball flattens slightly against the strings of your racket, storing potential energy.

Upon deformation, the ball experiences a transfer of energy from the racket, resulting in its acceleration.

Backhand Contact Point Biomechanics

5. Center of Percussion and Sweet Spot:

The center of percussion, also known as the sweet spot of the racket, is the point on your racket face where minimal vibration and shock are felt during ball impact.

Striking the ball near the sweet spot, allows for maximal energy transfer. This results in a more controlled and powerful shot.

6. Muscle Stretch-Shortening Cycle:

The contact point is a critical stage for the activation of the stretch-shortening cycle.

Rapid stretching of the muscles during the backswing phase stores elastic energy, which is subsequently released during the forward swing at the contact point, contributing to racket head speed.

Backhand Contact Point Biomechanics



7. Hip and Shoulder Rotation:

Correct hip and shoulder rotation during the contact point facilitates the generation of torque and power.

The separation of hips and shoulders allows you to load energy into your core and then efficiently transfer it to your racket for an explosive stroke.

Backhand Contact Point Biomechanics

8. Timing and Visual Focus:

Visual focus on the ball is crucial for optimal timing and accurate shot-making.

Fixating on the ball helps ensure that your racket meets the ball precisely at the intended contact point.



9. Follow-Through and Stroke Integrity:

After contact with the ball, the follow-through is essential for a powerful and consistent stroke.

A complete and smooth follow-through allows for the dissipation of excess energy, reducing the risk of injury and enhancing accuracy.

Backhand Contact Point Anatomy

1. Wrist and Forearm Muscles:

The muscles of your wrist and forearm play a crucial role at the contact point.

Wrist extensors and flexors stabilize your wrist and allow for precise control of the angle of your racket and control of the ball.

2. Shoulder Girdle and Rotator Cuff Muscles:

The shoulder girdle, comprising of the *scapula and clavicle*, provides a stable base for your upper arm during the stroke.

The rotator cuff muscles (*supraspinatus, infraspinatus, teres minor, and subscapularis*) work synergistically to control shoulder movements and provide dynamic



stability during the
contact.

Backhand Contact Point Anatomy

3. Upper Arm Muscles:

The muscles of your upper arm, such as the *biceps brachii* and *triceps brachii*, are active during the contact.

Your biceps contribute to racket acceleration, while your triceps help in decelerating your racket after impact.

4. Elbow Joint and Ligaments:

The elbow joint, formed by the articulation of the *humerus*, *radius*, and *ulna bones*, undergoes flexion during contact with the ball.

Ligaments, such as the *medial and lateral collateral ligaments*, provide stability to your joint during the forceful movements involved in the stroke.

Backhand Contact Point Anatomy

5. Ball of Foot and Toe Propulsion:

Proper weight transfer during the backhand involves the ball of your back foot. *(The ball of your foot is the padded area between the toes and the arch)*

Your toes contribute to propulsion and help maintain balance and stability during these rapid rotational movements.



6. Core Muscles:

Your core muscles, including your abdominals and obliques, provide stability. This helps you to transfer energy from your lower body to your upper body during the contact point.

A strong core significantly helps in creating an efficient kinetic chain.

Backhand Contact Point Anatomy

7. Cervical Spine (Neck) Muscles:

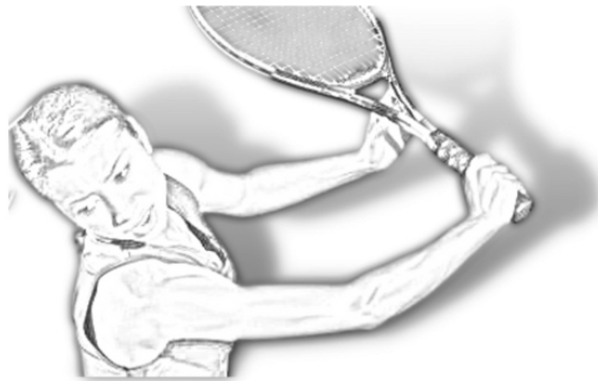
The muscles of your neck and cervical spine work to stabilize and maintain the correct head position during the stroke.

Precise head positioning helps you to focus on the ball and are critical for accurate ball striking.

8. Hips and Pelvis Alignment:

The alignment of your hips and pelvis is vital for optimal transfer of power.

Correct hip rotation and pelvic stability contribute to the generation of torque and angular momentum, enhancing the speed and control of the stroke.



Backhand Contact Point Anatomy

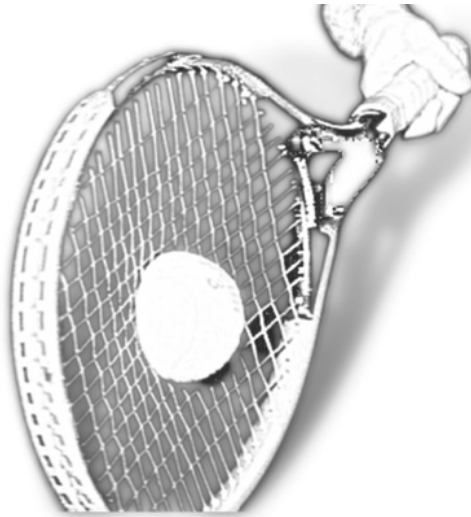
9. Racket Grip and Hand Positioning:

Your grip on the racket and hand positioning, significantly influence the outcome of your stroke.

Your grip type (Continental, Eastern, Western) affects the angle of your racket face at contact.

Hand pronation and supination contribute to different types of spins (topspin, slice or flat).

Good conditioning, technique refinement, and "bio-mechanical awareness" are key to developing a powerful, accurate, and consistent stroke.



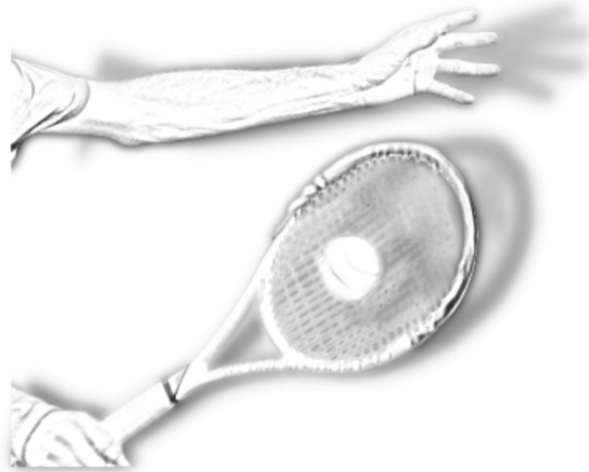
Backhand Contact Point Physics

1. Conservation of Momentum:

When the ball collides with your racket, the principle of conservation of momentum comes into play.

This law states that the total momentum of the system (in this case the ball and the racket) before the collision is equal to the total momentum after the collision.

As you swing your racket forward, it gains momentum, and upon contact with the ball, the momentum is transferred to the ball, propelling it in the opposite direction.



Backhand Contact Point Physics

2. The Coefficient of Restitution:

The *coefficient of restitution* (COR) is a measure of how much energy is conserved during the collision.

A perfectly elastic collision would have a COR of 1, meaning no energy is lost during the collision.

In tennis, the COR is usually slightly less than 1 due to energy dissipation as heat and sound.

A higher COR means more energy is transferred to the ball, resulting in a livelier shot.

Great ball strikers always have a COR much closer to 1 than normal players.



Backhand Contact Point Physics

3. Impact Time and Impulse:

The duration of time over which your racket and ball are in contact during the collision is known as the impact time.

A longer impact time leads to a smaller force applied to the ball, resulting in a slower stroke.

Conversely, a shorter impact time generates a larger force, leading to a faster stroke.

The change in momentum experienced by the ball, is referred to as impulse.

Impulse is equal to the force applied to the ball multiplied by the time that your racket and ball are in contact with each other.

Backhand Contact Point Physics



4. Racket String Tension and Surface Area:

The tension of your racket strings affects the amount of energy transferred to the ball during the collision.

Additionally, the surface area of the racket that comes into contact with the ball influences the amount of force applied.

A larger sweet spot on the racket face creates a more forgiving shot, even if the contact point is slightly off-center.

Backhand Contact Point Physics

5. Spin Generation:

The physics of the racket-ball collision is also responsible for generating different types of spin on the ball, such as topspin, backspin, and sidespin.

The angle of your racket face at the moment of impact, along with the direction and speed of your swing, determines the spin direction and rate of rotation.

6. Follow-Through and Ball Trajectory:

The follow-through after the racket-ball collision affects the trajectory and control of the stroke.

A smooth follow-through facilitates better control and accuracy, while an abrupt follow-through may result in a less predictable trajectory.

7. Air Resistance and Ball Flight:

As the ball leaves your racket, it encounters air resistance, which slows it down and influences its flight path.

Understanding the aerodynamics of the ball can help you anticipate its trajectory and reduce unforced errors

Backhand Contact Point Solution

The contact point on the backhand can be a problem for many players.

If the ball is too close to your body, then it's almost impossible to play a full drive.

If the ball is too close and you do not have enough space in front of your body, then it's also very difficult to achieve a powerful stroke.

You often see players simply slicing or chopping at the ball because their contact point is late and not far enough in front of their head and body.

A great way of finding a perfect contact point every time is to try to see the ball through your racket strings.



Seeing the ball through your racket strings will result in you naturally creating space between you and the ball.

Backhand Contact Point Solution

This space is essential as it creates a 'stable alignment' of your hand, wrist, and arm, through to your shoulder.

Stable alignments occur frequently in everyday life. If you are trying to push a heavy object, you will naturally position your legs, body and arms in a stable alignment.



It might be impossible to actually see the ball through the strings this way, but in striving and attempting to see the ball in this manner, you will create space between you and the ball.

You will have created a stable alignment and have a powerful and controlled stroke.

YouTube: <https://youtu.be/qCrs7JFTPWQ>

Backhand Power-Source Biomechanics

As you draw together your scapulae during the upswing of your backhand stroke, you engage large muscle groups, optimizing power generation.

1. Scapular Retraction and Adduction:

After your racket take back, focus on squeezing your shoulder blades together towards your spine (*scapular retraction*) and bringing your shoulder blades closer to each other (*scapular adduction*).

This action creates a stable and solid base for your upper body movement during your stroke.

YouTube: <https://youtu.be/NPLjNFylPho>



Backhand Power-Source Biomechanics



2. Muscle Activation Sequence:

Initiate the stroke, by engaging your *rhomboid muscles*.

Feel the muscles between your shoulder blades working to pull the *scapulae* medially.

Then, involve your *trapezius muscles*, which help elevate the *scapulae* and add to your *scapular adduction*.

Additionally, activate your *serratus anterior muscles* to stabilize the *scapulae* against your ribcage, preventing any winging (this is where the scapula protrudes from the back) and ensuring smooth scapular movement.

Backhand Power-Source Biomechanics

3. Core Stability and Lower Body Connection:

Remember that drawing your *scapulae* together is not just an upper-body movement.

Engage your core muscles, particularly the *Transverse abdominis* and *internal obliques*, to anchor your upper body to your lower body.

This stable core allows for the efficient transfer of force from your *gluteus maximus* and *hamstring muscles* in the lower body to your upper body during the upswing.



4. Kinetic Chain and Segmental Coordination:

Think about the kinetic chain as you perform the movement. Focus on the sequential activation of muscle groups, starting from your lower body, moving through your core, and reaching your upper body.

This segmental coordination ensures that energy is transferred optimally from the ground to your racket.

Backhand Power-Source Biomechanics

5. Shoulder Girdle and Glenohumeral Joint Stability:

Pay attention to your *shoulder girdle* stability as you draw together your *scapulae*. Feel the dynamic stability provided by the engagement of your *rhomboids*, *trapezius*, and *serratus anterior muscles*, supporting your *glenohumeral joint* during the forceful movement of your backhand stroke.



Backhand Power-Source Biomechanics

6. Scapulohumeral Rhythm:

Coordinate the movement between your scapulae and your upper arm bone (humerus) to establish scapulohumeral rhythm.

This coordinated movement, ensures that your shoulder joint functions optimally during the upswing.

7. Proprioceptive Feedback:

Be aware of the position of your shoulder blades and ensure that they are coming together at the correct time.

To achieve this, rely on *proprioceptive feedback* to sense the orientation of your shoulder blades throughout the upswing.

Understanding and applying the biomechanics of drawing together your *scapulae* during the upswing will enhance the stability and coordination of your upper body.

It will help you to optimize power and help you to achieve greater consistency.

Backhand Power-Source Anatomy

1. Rhomboid Muscles:

As you retract and squeeze your *scapulae* together, the *major and minor rhomboid muscles* play a significant role.

These muscles lie between your shoulder blades and are responsible for pulling the *scapulae* closer to the spine, providing stability and control during the upswing.

2. Trapezius Muscle:

The *trapezius muscle* is another crucial element in *scapular retraction*.

The upper fibers of the *trapezius* help elevate the *scapulae*, while the middle and lower fibers aid in *scapular adduction*, drawing them toward the spine and promoting the initiation of the upswing.

3. Latissimus Dorsi Muscle:

The *latissimus dorsi*, commonly known as the lats, contribute to *scapular adduction* during the upswing.

These broad muscles of the back help draw the *scapulae* medially, creating a stable platform and a powerful backhand.

Backhand Power-Source Anatomy

4. Pectoralis Major Muscle:

The *pectoralis major*, or pecs, also plays a role in *scapular retraction*.

These large chest muscles assist in pulling the *scapulae* together, working in conjunction with the *rhomboids* and *trapezius* to facilitate efficient upper-body movement.

5. Deltoid Muscles:

The posterior fibers of the *deltoid muscles* are activated during *scapular retraction*.

These muscles assist in shoulder joint stabilization, providing the *scapular* control needed for a coordinated upswing.



Backhand Power-Source Anatomy

6. Serratus Anterior Muscle:

The *serratus anterior muscle*, located along the sides of your ribcage, is instrumental in stabilizing the *scapula* against the ribcage during the upswing.

This muscle prevents "winging" of the *scapulae*, ensuring proper *scapular* movement and function. If your shoulder blades protrude from your back during the stroke, then you should exercise and develop the *serratus anterior muscles*.



Backhand Power-Source Anatomy

7. Transverse Abdominis and Internal Obliques:

Engagement of the *transverse abdominis* and *internal obliques* during your upswing provides core stability.

A stable core allows for the transfer of force from your lower body to your upper body. This contributes to the generation of power.



8. Gluteus Maximus and Hamstring Muscles:

The *gluteus maximus* and hamstring muscles contribute to the stability and force generation during the upswing.

The large muscles of your lower body, help to provide you with a solid base for the transfer of energy from the ground to your upper body.

Backhand Power-Source Physics

1. Newton's First Law of Motion (Law of Inertia):

Newton's first law states that an object at rest will remain at rest, and an object in motion will continue in a straight line at a constant velocity unless acted upon by an external force.

As you initiate the movement to draw your *scapulae* together, you must overcome the inertia of your shoulder blades and the surrounding muscles, to create the necessary acceleration.

2. Newton's Second Law of Motion (Law of Acceleration):

According to Newton's second law, the acceleration of an object is directly proportional to the force applied to it, and inversely proportional to its mass.

When drawing together your *scapulae*, you apply force through the activation of various muscles, such as the *rhomboids and trapezius*.

The acceleration of your *scapulae* depends on the magnitude of this force and the mass of the *scapulae* and surrounding tissues.

Backhand Power-Source Physics

3. Newton's Third Law of Motion (Action-Reaction):

Newton's third law states that for every action, there is an equal and opposite reaction.

As you exert force to draw your *scapulae* together, your *scapulae* exert an equal and opposite force back on the muscles involved in the movement.

This reaction allows for the coordination and stability necessary to execute your backhand.

4. Law of Conservation of Angular Momentum:

The movement of drawing your *scapulae* together involves angular momentum.

When you retract and adduct your *scapulae*, you change the distribution of mass around the axis of rotation (shoulder joint).

The conservation of angular momentum dictates that as you decrease the distance between your *scapulae* and the axis of rotation.

This means that the angular velocity increases, aiding in the smoothness and efficiency of the stroke.

Backhand Power-Source Physics



5. Law of Levers:

The movement of drawing your *scapulae* together can be understood through the principles of levers in biomechanics.

Your *scapulae* act as a lever arm, and the force you apply (via muscle activation) at the insertion points creates the desired movement at the fulcrum (shoulder joint).

Backhand Power-Source Physics

6. Law of Equilibrium and Stability:

Biomechanical principles related to equilibrium and stability are vital during this movement.

To maintain stability, the forces exerted on your *scapulae* must be balanced and coordinated, ensuring that your shoulder girdle remains steady throughout your upswing.

Backhand Power-Source Physics

7. Law of Energy Conversion:

The movement of drawing together your *scapulae* involves the conversion of muscular energy into kinetic energy.

The force generated by your muscles transfers the kinetic energy to your *scapulae*, allowing for the desired motion during your backhand.



Backhand Power-Source Solution

The Secret Power Source

Do you appear to have good technique on your backhand and yet struggle to hit with power?

You can often redirect the ball if it's coming to you quickly but generating power and heavy topspin remains a mystery to you? Sound familiar?

The primary reason you may struggle to create power is that you're pushing forwards into the stroke and making contact with the ball using the lateral side of your arm.

Unfortunately, the lateral (external) side of your arm is fundamentally weak, much weaker than the medial (internal) side.

The lateral side refers to the outer arm from the elbow outwards, and the medial side is the inner arm from the elbow inwards.

When you push from the outside of your arm, there's no real major muscle group to help you attain a strong, stable, and fast stroke.

Backhand Power-Source Solution

As you take your racket back, push your *Rhomboid* muscles apart. These are the big muscles in your back.



Backhand Power-Source Solution

Stretching your back in this way gathers energy which you can then reverse, creating a much more powerful stroke



Backhand Power-Source Solution

Now, reverse this movement.



Draw your shoulder blades (the *scapulae*) together at the exact moment that you begin the swing.



This way, your *pectoral muscles* and *deltoids* will project your arms away from your body.

Backhand Power-Source Solution

This is a very stable and strong movement, which can be executed with much more force and speed than when you only use your arm.

The use of the whole body and the major muscle groups is why you see both arms projected away from the body when watching professional players hit the backhand stroke.



By using your shoulder blades this way, your backhand becomes a throwing action, instead of a pushing action. This generates much more speed, stability, power and spin.

Backhand Power-Source Solution

So, don't push your arm and racket away from your body.

Ensure that when you want to generate power, you draw your shoulder blades together and project your arm and racket away and up.

You will utilize your body's major muscle groups and have as table, powerful, and consistent backhand.

YouTube: <https://youtu.be/7lggbCg4U74> and https://youtu.be/1n8qX2Fe_-8

Feel Good and Funny Stories

Some of you may know that one of the biggest problems in tennis is the incredibly high percentage of high level junior players who quit the game.

It is a much higher number than in any other sport. In the USA about 70% of tournament level juniors quit by the age of 16, as a comparison, in Basketball the number is about 10%.

I think that the reasons for this are fairly obvious.

Tennis is an individual sport. There are no teammates to blame. There are no draws, you either win or lose.

If 32 young Roger Federer's enter a tournament, then 31 go home as losers.



Feel Good and Funny Stories

Most tournaments are one to two hours from home, so a mentally and physically exhausted young player is trapped for a long time with a well-meaning but often misguided parent. To get home they must endure what becomes "the dreaded car journey"



The journey almost always takes this format.
20 minutes of silence.

Then it begins, with well-meaning but ultimately depressing remarks.

"I don't mind when you lose, it is when you do not play your best tennis" Is usually the opener.

The young player is trapped and must suffer hours of this unintentional abuse with the recent match being forensically examined over and over again.

After several years of this most young players quit.

Feel Good and Funny Stories

The dropout rate from my many years as a coach was almost zero.

This was because we had an agreement, a pact between myself, the parents and the player.

If the player could answer yes to two questions, then nobody could talk about the match for 24 hours.

The two questions were;

"Did you fight for every point?"

"Did you behave with dignity?"

When we teach tennis, we teach more than tennis!



Feel Good and Funny Stories

What more can we give a young person than the skill and gift of fighting for everything and maintaining discipline in high-pressure situations?

One single ball that is not chased.

One single throw of a racket or bad word and the answer to either of the questions is not yes.

The pact is therefore broken and the match can be discussed endlessly.

Players who learn to fight for everything and behave with dignity may become successful tennis players. But with these essential skills they WILL become successful human beings.



STEPHEN JAMES BOURNE

I have been involved in tennis all of my life.

As a player I was a good hitter, I was ranked in the top 30 players in the UK, often helping professional players prepare for matches.

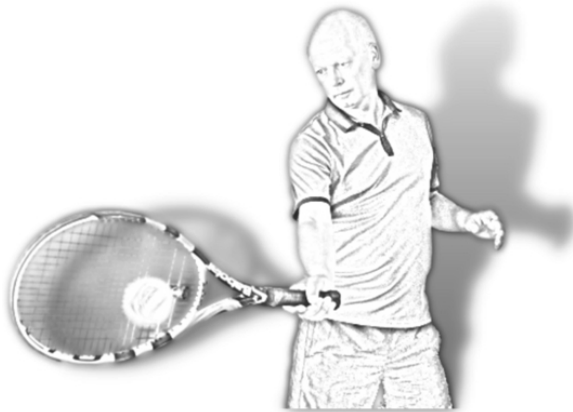
Having previously coached in the USA, Arabia and the UK I ran my own tennis academy for many years on the Eastern coast of Spain.

For over 20 years we worked with every imaginable type of player.

Beginner to professional, young and old (my oldest client was 93 years of age)

Aided by a large team of coaches I developed a unique approach to teaching
(see The One Minute System on page 3)

This system proved to be amazingly successful at all levels.



STEPHEN JAMES BOURNE

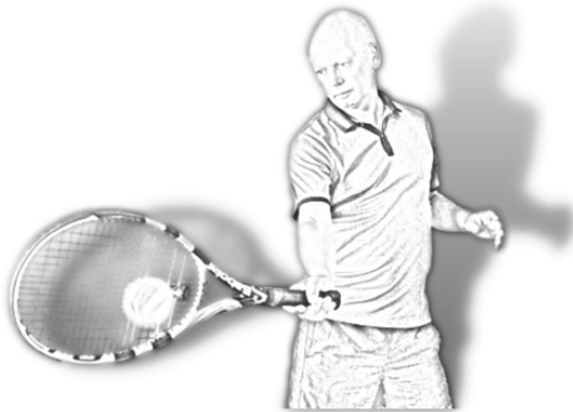
During this time both recreational and competitive players requested online help and written information on the teaching methods that they had experienced and benefitted from. The pandemic gave me the opportunity to take this teaching philosophy online and the One Minute Tennis YouTube channel began.

The channel enjoyed amazing success with over 2,000,000 views in the first year.

<https://www.youtube.com/channel/UCbmBx0-ZS9V7KSbyhrZcYqw>

This series of Tennis Solutions Books aims to help recreational and developing players use the techniques of the best players in the world.

The combination of the scientific principles of the latest tennis techniques and my quick and easy-to-follow One Minute Solutions make a complicated game simple and will help you to play better tennis today!



GET YOUR ONE MINUTE TENNIS SOLUTION BOOKS NOW

